Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building and supporting **SQL-backed enterprise applications**, **web APIs**, and cloud workflows, with experience in testing, documentation, design reviews, and production issue resolution.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized PostgreSQL and SQL Server-backed application workflows supporting 200+ users and 10K+ structured records, reducing latency by **35%** through query tuning, indexing, and validation improvements.
- Built ETL-style integration pipelines processing 5,000+ monthly jobs with reconciliation checks, error handling, and CI/CD-backed releases, improving production reliability and speeding support for data and runtime issues.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed full-stack web features and REST APIs handling 10,000+ daily requests, translating product needs into maintainable backend services and interfaces while participating in testing, debugging, and iterative releases.
- Implemented secure batch-processing workflows for PDF, DOCX, and ZIP ingestion with AWS-backed persistence, improving processing reliability and reducing feedback turnaround by **60%** in production.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based evaluation pipelines across 100,000+ records, creating reproducible reporting and test scenarios that supported system analysis, debugging, and clearer communication of technical tradeoffs.
- Produced technical reports and targeted validation scenarios translating findings into actionable guidance, improving collaboration, reducing handoff ambiguity, and supporting better design-review discussions across teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with microservice architecture, automated VM testing, and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across collaborative Agile delivery cycles.
- Troubleshot and optimized API integrations using OOP design patterns, AWS Glue, and S3, improving response times by **30%** and increasing pipeline efficiency by **68%** across production workloads.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based resource management platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six core business workflows through modular, maintainable application components.
- Deployed Azure SQL with Jenkins-based CI/CD and automated testing, achieving 95%+ successful build validation while improving release consistency and production-readiness across delivery cycles.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++, Bash
Application Development: REST APIs, Flask, FastAPI, Node.js, React, Next.js, object-oriented design, system design
Databases & Reporting: PostgreSQL, MySQL, SQL Server, Oracle familiarity, MongoDB, relational schema design, query tuning
Delivery & Support: AWS, Azure, Docker, Jenkins, Git/GitHub, CI/CD, automated testing, secure development practices

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL development workspace supporting 10+ databases, accelerating schema exploration, query authoring, and data management across Java and TypeScript modules in enterprise-style environments.

HMIS Code | *MERN, TypeScript, ABDM APIs, Sentry*
- Built a microservice-based healthcare management platform supporting 100+ daily appointment workflows, reducing patient-history retrieval by 2x through structured schema design, monitoring, and maintainable service boundaries.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a containerized RESTful data processing API on AWS with automated CI/CD releases and validation checks, improving deployment reliability, interoperability, and supportability for production data workflows.